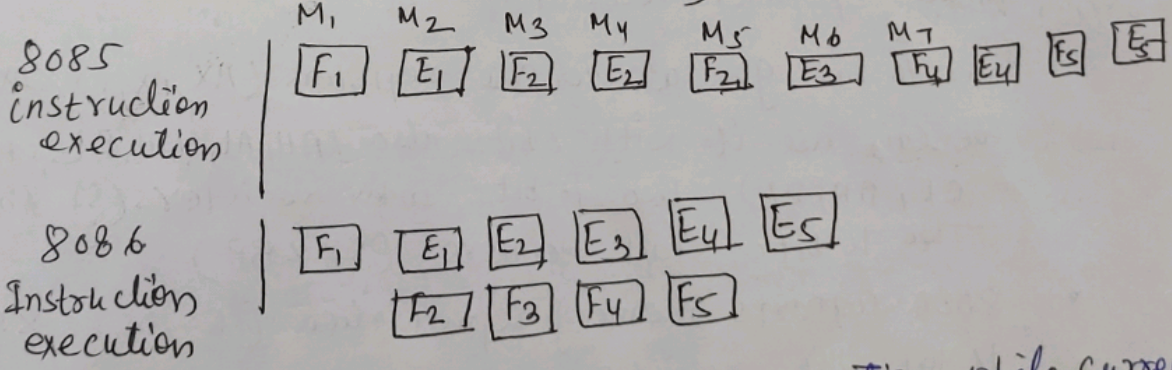
* Feature of 8086 µP

It is upgraded µP from 8085. 8086 has following features

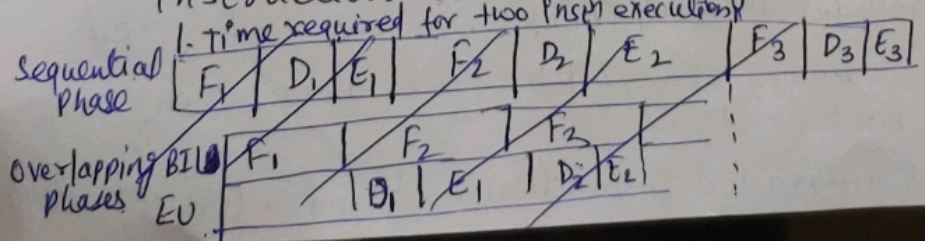
- 8086 has 16 bits ALU. [older version 8085 has 8 bits ALU] so, operating speed 8086 have increased significantly well.
- 8086 has 16 bits of Data bus. So in one machine cycle it can transfer 2 bytes. [older version 8085 has 8 bits of data bus]
- 8086 has 20 bits of Address bus, with 8086 we can interface total memory by 2^{20} = 1MB (older version 8085 has 16 bits Address bus)
- 8086 has two internal hardware unit. BIU - Bus interface unit & EU - Execution unit.
- 8086 supports pipelining. So multiple instructions can be executed in parallel to increase execution speed of microprocessor.
- Four 16 bit general purpose registers (AX, BX, CX & DX). We can use it with 8 bits also (AH, AL, BH, BL, CH, CL, DH, DL). Two 16 bits Index registers (SI & DI). Two 16 bits stack pointers (SP & BP)
- 8086 supports memory segmentation (CS, DS, ES & SS)
- 16 bit flag register.
- Two hardware modes: Maximum mode & Minimum mode
- 8086 provides memory banks. So 1MB is bisected into two bank of 512 KB, Lower bank (even) and higher bank (odd).
- 8086 has upgraded instructions set and interrupt Service

Pipelining in microprocessor 8086

* How to increase speed of execution of programs in μP.
  - Increase data bus [so, in single machine cycle we can transfer more data]
  - Increasing system clock [so, more cycle can get executed in given time]
  - Upgrading ALU [16 bits ALU perform more arithmatic & logical operations compared to 8bits of ALU)
  - Upgrading Memory technology : [like nowadays (2022), we have SSD & Cache in our memory]
  - Upgrading instruction set. (like in 8086 we have Multiplication & Division instruction available which was not available with 8085)
  - Pipelining (it helps to execute multiple instruction in parallel)



→ Features of fetching the next instruction while current instruction is executing is called Pipelining.



* Diagram is given in next page.

* issues of piplining in microprocessor 8086.

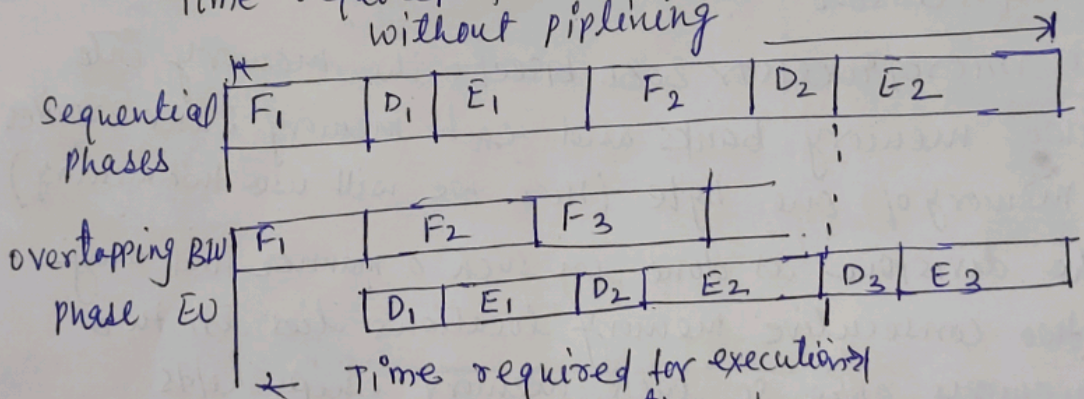
- In branch instruction execution, next fetch instruction will be discarded.

  i.e. In case of Jump & CALL instructions, instruction already fetched in queue are of no use, in these case queue is dumped and newly formed by loading instruction from new address specified by Jump or call instruction.

8086 instruction Execution	F_1	E_1	F_{2B}	E_{2B}	F_3	E_3	F_4	E_4

(Note- but in current, branch predicing algorithm is dereloped by Intel. so this types of issues has been resolved.)

Time required for execution of two instruction without piplining



Memory Banking of 8086

Since 8086 is having 16 bits data bus so 8086 wants to exchange to total 2-bytes in one machine cycle. But one memory location is ~~store~~ having 1 byte data so that issue is resolved in 8086 by memory banking.

- 8086 has 16 bits of data lines, so its should access 16 bits (2 bytes) in one machine cycle.
- To perform 2 bytes data transfer, so it need to read 2 memory locations as one memory location has only one byte of data.
- If both of these memory locations are there is same memory then it can not be accessed by microprocessor.
- So microprocessor 8086 bisects the memory into two memory banks and each memory bank provides memory of one byte. (Here we will use two memory)
- The division is done in such a manner that any two consecutive memory locations lies in two memory chip so each memory chips holds alternate memory locations.
- Hence one memory bank have all even address (Even memory bank) and other memory bank have all odd address (Odd memory bank).

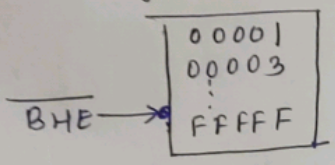
* Generally for any 16 bits operation, Even memory bank provides the lower byte (Lower Bank) and Odd memory Bank provides Higher byte (Higher Bank). So we should write Prog from Even memory locations.

Memory Banking in 8086

1MB Memory [A19-A0]

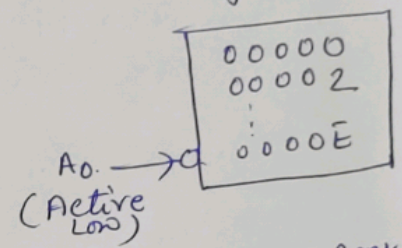
512 KB (ODD BANK)

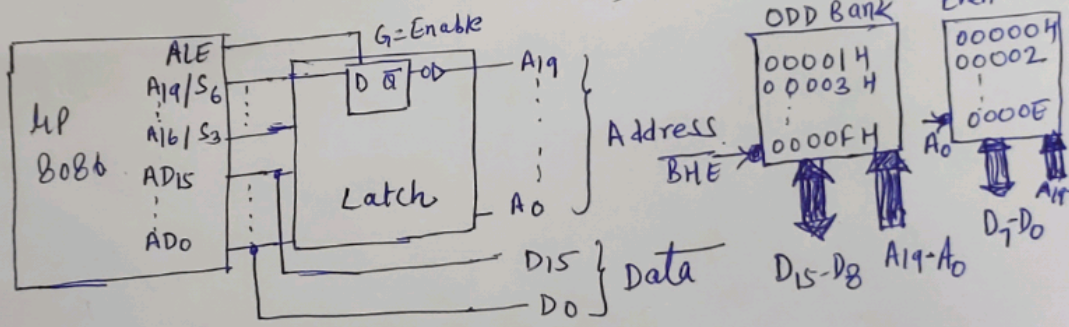
- It is also called higher Bank.
- These memory bank is selected when $\overline{BHE}$ = 0
- Range of Address is



512 KB (Even Bank)

- It is also called Lower Bank
- These memory bank is selected when A0 = 0
- Range of Address is

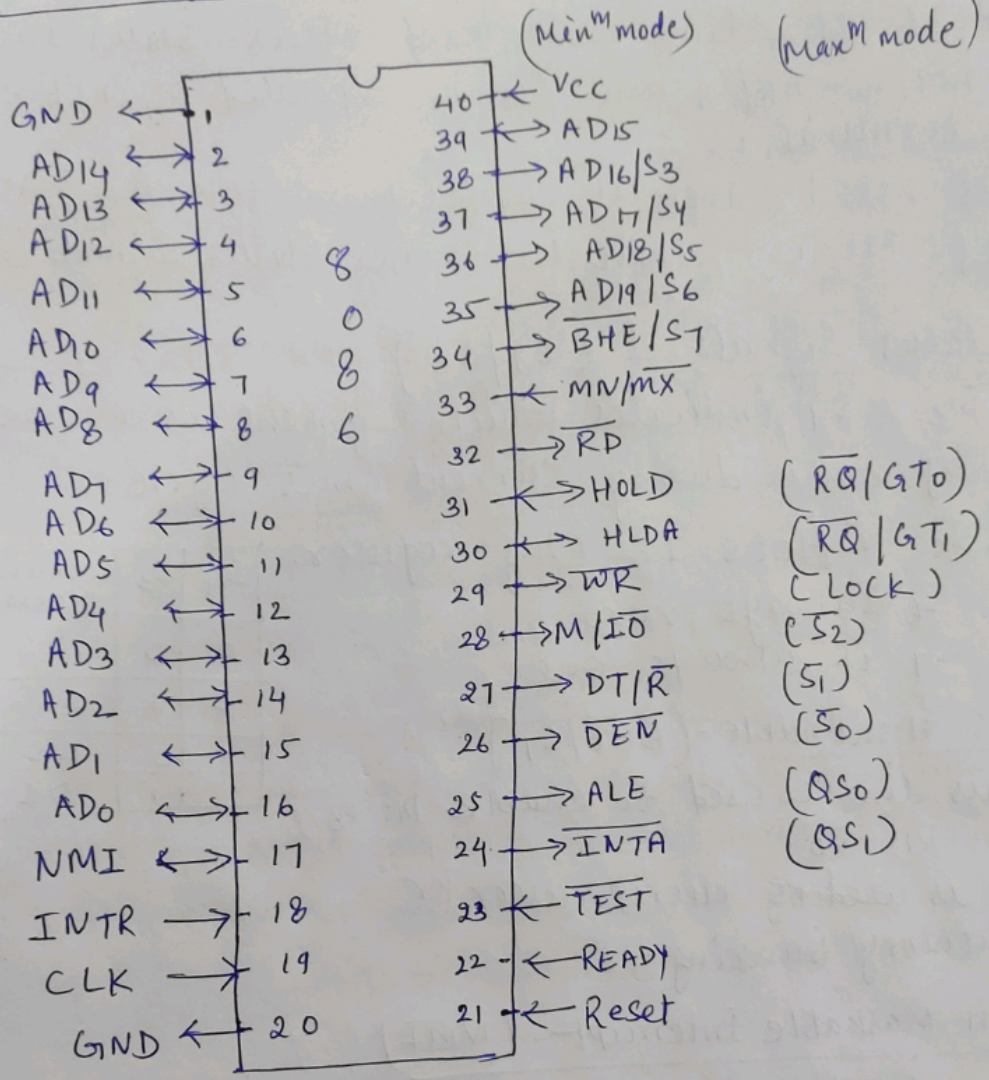




A0	$\overline{BHE}$	Machine Cycle	cycle Data	Operation
0	0	1	D15-D0	Read a word from Even Add
0	1	1	D7-D0	Read a byte from Even Add
1	0	1	D15-D8	Read a byte from odd add
1	0	2	D15-D8	Read word from odd Add
0	1		D7-D0	

Note: To Read word for 1 machine cycle, the Address should be even. If Address is odd, then need two machine cycle for read word.

Pin Diagram of microprocessor 8086



Address Data bus – [AD0-AD15]

- 8086 has 20 lines for Address bus and 16 data line for data bus.
- Here AD0-AD15 are time multiplexed Address Data line and can be seperated by ALE terminal.
- If ALE = 1 then it carries Address [A0-A15] and if ALE = 0 then it carries data [D0-D15].

- **Address status lines** – [A19/S6 – A16/S3]
  - 8086 has 4 time multiplexed Address status lines A19/S6 – A16/S3 and can be seperated by ALE terminal.
  - If ALE=1, then it carries Address [A19-A16] and If ALE=0, then it carries Status Signals [S6-S3]

- **Status Signals** — [S7/S3]
  - S_3 & S_4 indicates which segment is accessed by 8086 during current bus cycle.
  - S_5 reflets IF Flag register.
  - S_6 is always zero.
  - S_7 is always one.

S_4	S_3	Segment Accessed
0	0	ES
0	1	SS
1	0	CS
1	1	DS

- **Bus HighEnable** – ($\overline{BHE}$/S7)
  - This line is used to enable D15-D8 of D15-D0.
  - It is used by Microprocessor for memory banking.

- **Non Maskable Interrupt** – [NMI]
  - This line is used give highest priority Interrupt to microprocessor 8086.
  - It can not be disabled by the software.
  - It is positive edge triggered interrupt.

- **Interrupt Request** – [INTR]
  - This is level triggered hardware interrupt
  - It depends of status of Interrupt flag

- If IF=1, INTR=1, then 8086 get interrupted.
- If IF=0, INTR=1, then INTR is disabled.

Interrupt Aeknowledgment - ($\overline{INTA}$)
- Along with Interrupt request on INTR microprocessor gives interrupt acknowledgment to their peripherals.

System clock - [CLK]
- clock is given to 8086 for internal timining.
- Clock with 8086 is 5MHZ, 8MHZ & 10 MHZ.
- clock frequency depends on the version of 8086

8086 → 5MHZ
8086-2 → 8 MHZ
8086-1 → 10 MHZ.

Minimum & Maximum Mode [MN/$\overline{MX}$]
- For minimum mode, connected with VCC
- with minimum mode, 8086 will work as single microprocessor
- For maximum mode, connected with ground
- with maximum mode 8086 will works with multiple coprocessor.

READY Signal (Ready)
- If it is logic 0, 8086 enters into the wait state like idle.
- It is used to synchronizes low peripheral device.
- If ready pin is at logic 1, it has no effect on the operation.

Read ($\overline{RD}$) —
- If is at logic 0, microprocessor read data from memory or I/O devices.

System Reset - (RESET)
- This input will reset microprocessor 8086.
- If it ~~must~~ held logic 1 for minimum of 4 clock cycles then microprocessor will get RESET.
- After RESET of 8086, CS & IP initialized to FFFFH & 0000H, so physical address will be FFFF0H.
- It clears PSW, DS, SS, ES & instruction queue.

Test ($\overline{TEST}$)
- It is used for synchronization.
- When this input is logic 0, 8086 executes wait instruction.
- The 8086 enters into wait states after execution of WAIT instructruction until a low signal on the ~~Test~~ $\overline{TEST}$ pin.

Power Supply (Vcc & ground)
- 8086 is given with 5 volt DC supply
- 8086 has two ground pins.
- This two ground are there to have less power dissipation

Minimum mode of 8086 (24 to 31 pins signals)

INTA - (Interrupt Acknowledge output)

- This indicates recognition of an interrupt request.

ALE - This signal is provided by 8086 to demultiplexed the AD0-AD15 into A0-A5 and D0-D15 using external latches.

DEN (Data enable) output: ~~This signal is used to control data flow direction.~~

- This signal informs the transceivers that CPU is ready to send or receive data.

DT/$\overline{R}$ (Data transmit/Receive) output:

- This signal is used to control data flow direction.

  DT/$\overline{R}$ = 1 (high) then 8086 is transmitting the data

  = 0 (low) then 8086 is receiving the data

M/$\overline{IO}$ output:

- It is used to distinguish memory data transfer (M/$\overline{IO}$ = high) and I/O data transfer (M/$\overline{IO}$ = low).

$\overline{WR}$: (Write output) - $\overline{WR}$ is low whenever the 8086 is writing data into memory or an I/O device.

HOLD (Input), HLDA (output): A High on Hold pin indicates that another master (DMA) is requesting to take over the system bus.

- On receiving HOLD signal processor sends HLDA signal High as an acknowledgement.

Maximum Mode

1. QS_1, QS_0 (output):
   - These two output signals reflect the status of instruction queue.
   - This status indicates the activity in queue.

QS_1	QS_0	Status
0	0	No operation
0	1	First byte of an opcode
1	0	Queue is empty
1	1	Subsequent byte of an opcode.

2. $\overline{S_2}$, $\overline{S_1}$, $\overline{S_0}$ (output):
   - These three status signal indicate the type of transfer to be take place during the current bus cycle.

$\overline{S_2}$	$\overline{S_1}$	$\overline{S_0}$	Machine cycle
0	0	0	Interrupt acknowledge
0	0	1	I/O Read
0	1	0	I/O write
0	1	1	Halt
1	0	0	Instruction fetch
1	0	1	Memory Read
1	1	0	Memory write
1	1	1	Inactive - Passive.

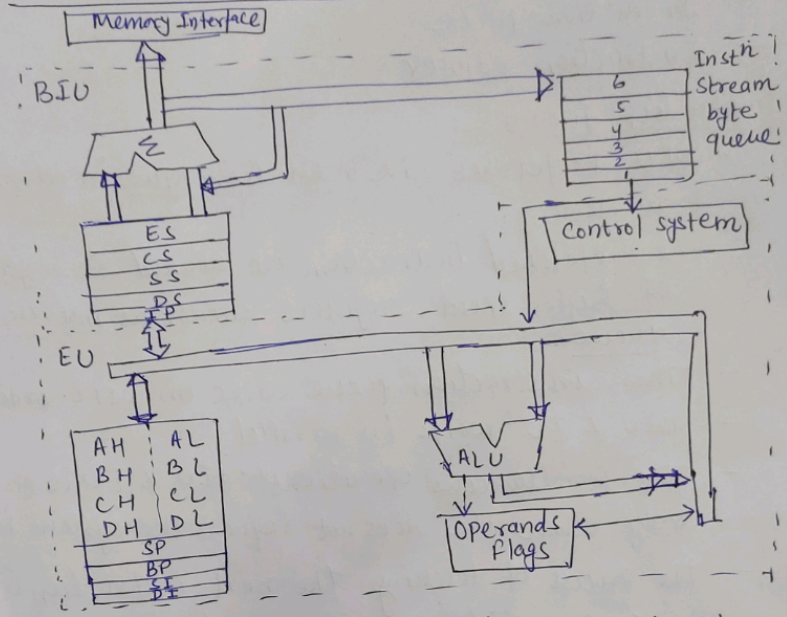
$\overline{LOCK}$:

- This signal indicates that an instruction with a Lock prefix is being executed and bus is not to be used by another processor.

$\overline{RQ}/\overline{GT_1}$ and $\overline{RQ}/\overline{GT_0}$:

- In max^m mode, HOLD & HLDA pins are Replaced by $\overline{RQ}$ (Bus request) / $\overline{GT_0}$ (Bus Grant) and $\overline{RQ}/\overline{GT_1}$ signals.
- By using bus request signal another master can request for the system bus & processor Communicate that the request is granted to the requesting master by using bus grant signal.
- Both Signals are similar except the $\overline{RQ}/\overline{GT_0}$ has higher priority than $\overline{RQ}/\overline{GT}$.

Architecture of Microprocessor 8086



* BIU is responsible for establishing communications with external peripheral devices and memory via system bus.
* Main functions of BIU is followed
  - It fetches the instruction from memory
  - It reads data from I/O and memory.
  - It writes data into I/O and memory.
  - It provides the address relocation facility.
  - It sends the address of the memory or I/O
  - It supports Instruction queuing.

* BIU contains three main parts
  - segment registers
  - Instruction queue
  - Instruction pointer

• Instruction queue —
  - BIU prefetches six instruction bytes in advance from memory.
  - The prefetched instructions are stored in a group of high speed registers known as instruction queue.
  - This instruction queue works on FIFO order.
  - BIU & EU works in parallel.
  - The simultaneous operation of BIU & EU is possible only when the does not require the system bus.
  - The process of fetching the next instruction in advance while EU is executing the current instruction is pipelining.

• Segment Registers
  - 8086 µp has the capability of addressing 1MB memory which is divided into segments.
  - Each segment contains 64KB memory.
  - But any instant 8086 works with only four 64KB segments.

* Each segment associated with segment register
  - code segment register (CS) (16 bits)
  - Data segment Register (DS) - 16 bits
  - Stack segment Register (SS) - 16 bits
  - Extra segment Register (ES) - 16 bits
* Segment register of BIU is used to store the starting address of Memory Segment.
* BIU generates 20 bits address using segment register and offset pointer registers.

PA = SRX10H + OP

* Instruction pointer
  - IP holds the address of next instruction which is to be executed next.
  - It contains the offset value of next ~~instruction~~ address.

- Execution Unit of Microprocess 8086

EU informs BIU from where the next instruction or data to be fetched.

* EU performs following functions
  - It pickup the instruction from the instruction queue of BIU.
  - It decode the instructions & then executes the instruction
  - It updates the status of flag register.

△ Control unit

- It is controlling and coordinating all the activities of sub units
- It fetches, decodes and executes instructions.
- It gives control signal ~~signo~~ like read, write etc.

△ ALU of EU

- It performs all the airthmatic and logical operations.
- Results may be stored in general purpose register or Index register.
- It updates flag register after every instructions.

△ General Purpose registers & Pointer of EU

- EU have four general purpose registers. (AX, BX, CX & DX)
- These registers are used for accessing data very fast.
- EU hav SP & BP for stack.
- Index register are SI & DI for data & Extra segment respectively.

Register Organization (Programming model of 8086)

AX	AH(8)	AL(8)	Accumulator
BX	BH(8)	BL(8)	Base
CX	CH(8)	CL(8)	Counter
DX	DH(8)	DL(8)	Data

(a) General purpose Register

ES(16)	Extra segment
CS(16)	Code segment
DS(16)	Data segment
SS(16)	Stack segment

(b) Segment Registers

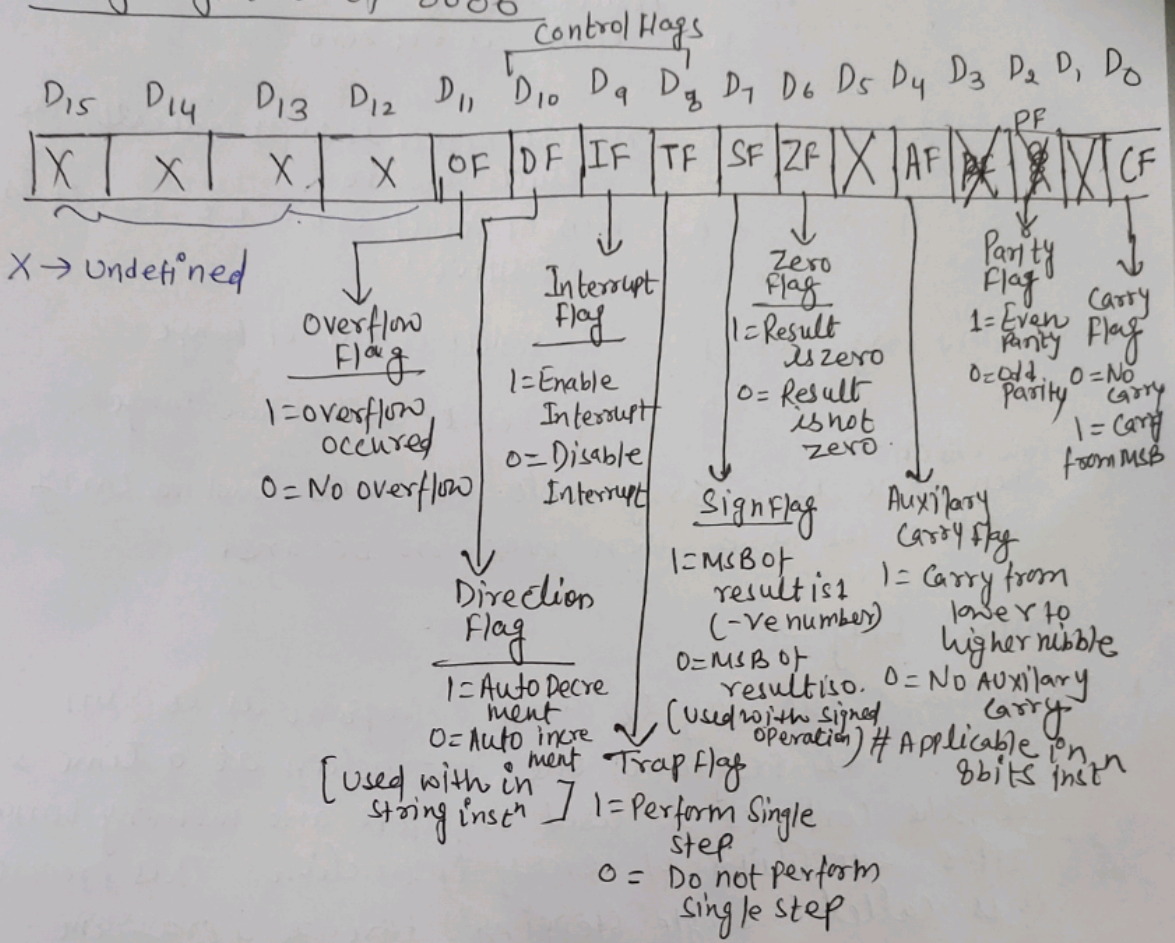
SP(16)	Stack pointer
BP(16)	Base pointer
SI(16)	Source Index
DI(16)	Destination Index
IP(16)	Instruction Pointer

(c) Segment offset pointers.

Flag(16bit)

Program status word

Flag register of 8086



→ Conditional flags

- Carry Flag (CF) — CF = 0 no carry out from msB
  CF = 1 carry out from msB
- Parity Flag (PF) PF = 1 Even Parity i.e no. of one's even
  = 0 odd parity i.e no. of one's odd
- Auxilary Carry (AF) AF = 1 Carry from lower nibble to Higher nibble (D_3 bit to D_4 bit)
  = 0 No Auxillary carry

- Zero Flag : ZF=1 Result is zero
  =0 Result is not zero.
- Sign Flag : SF=1 MSB of Result is 1. It indicates the result is -ve.
  =0 MSB of Result is 0. i.e result is positive.
- overflow Flag OF=1 if Result is out of Range.
  =0 if Result is within Range.

overflow check :-
When there is a carry into the MSB and no carry out of the MSB then overflow occurs.

Control Flag –

- Trap Flag : One way to debug a program is to run the program one instruction at a time & see the contents of used registers and memory variable after execution of every instruction. This processes is called 'Single stepping' through a program.

  TF=1 Perform ~~operation in~~ single step
  =0 Do not perform single step.
- Interrupt flag(If) : It is used to allow the interruption of a program. If set, a certain type of interrupt (a maskable interrupt) can be recognized by 8086 ; otherwise these interrupts are ignored.
  If = 1 Enable Interrupt
  = 0 Disable Interrupt
- Direction flag(DF) : It is used with string instructions.

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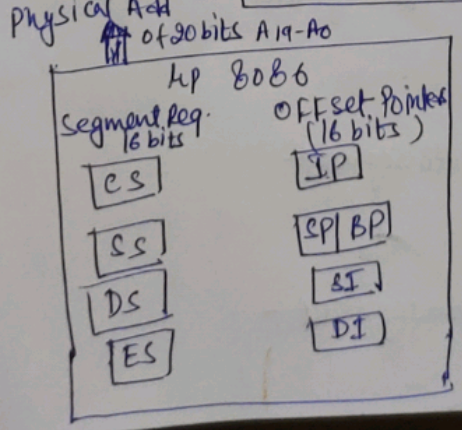
Memory Segmentation of 8086 –

* 8086 has 20 bits of physical address
  - Here 20 bits of physical address is not byte compatible physical address.
  - To avoid the issue of byte compatibility, microprocessor 8086 has uses memory segmentation. In which we uses concept of virtual address.
  - Here, memory is bisected in Four segments.
    - Code segment
    - Stack segment
    - Data segment
    - Extra Segment
  - To calculate Physical address from Virtual address we should know segment address and offset address.
  - Segment Address is held by segment registers (16 bits). [CS, SS, DS, ES]
  - OFFSET Address is held by OFFSET pointers (16 bits). [IP, (SP & BP), SI & DI]
  - So, Physical Address is calculated by

$$PA = SR \times 10H + OP$$



- SP (Stack Pointer) – SP points top of stack segment
- BP (Base pointer) – BP points any address of stack segment.

eg. If code segment register is 1245 H and instruction pointer is 1561 H. Then find physical address of given instruction

$$PA = SR \times 10H + OP$$
$$= 1245 \times 10H + 1561H$$
$$= 12450H + 1561$$
$$PA = 13981H$$

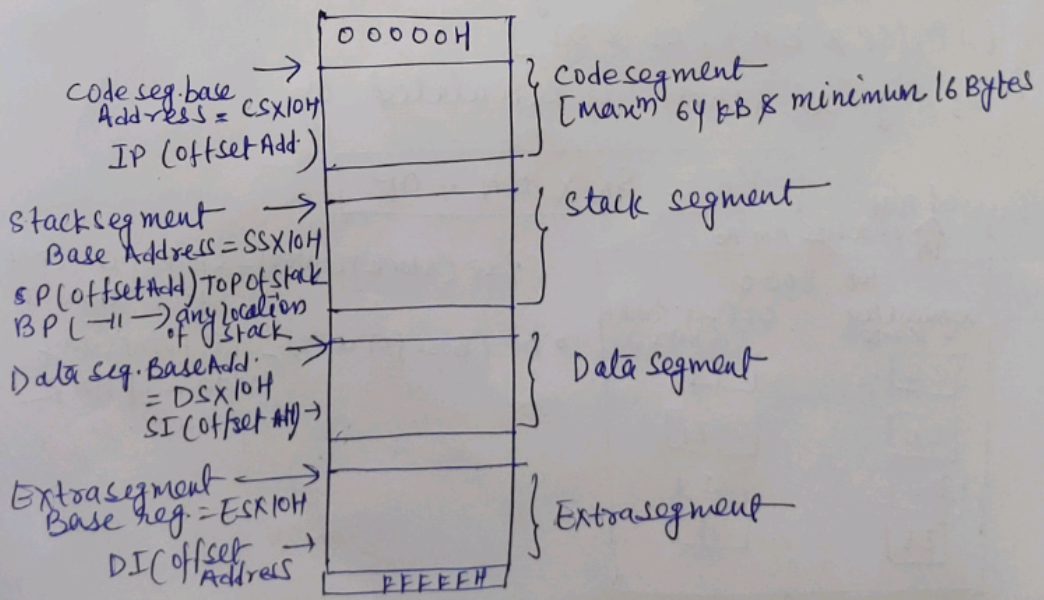
eg. 2. : If Data segment register is 78EF H. Then finds the maximum & minimum range of physical Address.

For Maxm Range.

$$PA = SR \times 10H + OP$$
$$= 78EF \times 10 + (0000H \text{ to } FFFFH)$$
$$PA = 78EF0 + 88EEFH$$

For minimum range (16 byte)

$$PA = SR \times 10H + OP$$
$$= 78EF0 + (0000H \text{ to } 000F)$$
$$= 78EF0 - 78EFF$$



dvantages of Memory Segmentation

- It allows the memory addressing capacity to be 1MB even though the address associated with individual instruction is only 16 bits.
- It facilitates seperate memory area for program, data and stack.
- It permits a program or its data to be put in different areas of memory, each time the prog. is executed i.e program can be relocated which is very useful in multiprogramming.
- change in one segment does not effect the other segment.
- It avoids memory overlap with different segments
- Memory management gets easier.
- Debugging gets easier.

8086 Question Bank

→ Write four differences bet^n 8085 & 8086 microprocessor? (4marks) (2021-22)

→ Explain the concept of memory segmentation in 8086 microprocessor with suitable diagram. What are the advantages of memory segmentation in 8086 microprocessor? (4marks) (2021-22)

→ Explain all control flags of 8086 µP. (3marks) (2021-22)

→ Why 8086 µP is faster than microprocessor 8085? How pipelining is performed in 8086 µP. Explain with help of diagram? (3marks) (2021-22)

→ Explain function of following 8086 µP pins. (2021-22) (5marks)
(a) LOCK' (b) TEST' (c) BHE' (d) S_0' S_1' S_2' (e) QS_1, QS_0.

→ Draw pin diagram of 8086 and Explain name of signal associated with max^m mode operation of 8086 (5marks) (2018-19)

~~Time~~

Time delay — Calculate time delay of given set of instructions. Assume the microprocessor is operating with the freq. of 2MHz.

```
        MVI B, 20H
LOOP2:  MVI C, FEH
LOOP1:  DCR C
        JNZ LOOP1
        DCR B
        JNZ LOOP2.
```

→ How many times the loop will be executed (write in decimal)? Find out total-T state for each loop. (2018-19)

(a)
```
    MVI B, 11H
L1: DCR B
    JNZ L1
```

(b)
```
    LXI H 1000H
L1: DCX H
    MOV A, L
    ORA H
    JNZ L1
```

(c)
```
    MVI C, 00H
L1: DCR C
    JNC L1
```

→ 8086 can generate 16 bits I/O Address. Hence it can access $2^{16} = 65536$ I/O ports

→ 8086 has multiplexed address & data bus which reduces the number of pins

Question Bank

→ PIC 8259

- Use schematic diagram to discuss the internal architecture of PIC 8259 ? (2018-19), (2021-22)
- How many 8259, s Ic can be connected in cascaded mode Show their Cascading structure?
- with proper required format explain the various operational command word of 8259 PIC.
- Draw the block diagram of Intel 8259 PIC. Also gives the initialization sequence of the 8259 ICW registers. (2017-18) (3 marks)

Ic 8237 –

1. Explain the data transfer types supported by 8237 and also explain the current & base register in 8237 ? (3 marks) (2018-19)
2. Discuss the command and mode register configuration of 8237 DMA controller. (2 marks) (2018-19)
3. What are the different I/O modes of transfer explain each one? How DMA removed the disadvanges with previous scheme? (2017-18) (3 marks)
4. Draw and explain the bit pattern of mode register of 8237A.

Question Bank

8255 – Draw Internal architecture of 8255. List the operating modes of 8255. Specify the bit of control word for 8255, which differentiate both I/O mode & BSR mode. (2021-22) SM. (2018-19) GLA (17-18)

2. Find the data direction and mode of operation of the 8255 ports, if control word written into it is 0AH. (2018-19)
3. Explain all three I/o modes of operation in IC 8255 and also gives some application of 8255? (2017-18) (3mark)
4. write a ALP to blink port C bit C^3 with 1 second delay of IC 8255. (2017-18)

8253 IC –

→ with the help of timing diagram explain mode 0 & mode 1 operation of 8253 timer/counter (3mark) (2018-19)

→ ~~Explain~~ Draw & explain the timing diagram of mode 2 operation of 8253 timer/counter, also explain the different mode of 8253. (2017-18)